

Discrete Vector Bundles with Connection

The combinatorial theory above sheaf neural networks, connection Laplacians and gauge-equivariant CNNs: **curvature, the Bianchi identity, products**

Daniel Berwick-Evans ■ Anil N. Hirani (presenting) ■ Mark D. Schubel

Department of Mathematics, University of Illinois Urbana-Champaign ■ hirani@illinois.edu ■ hirani.github.io



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I. THE BRIDGE what you already build

1 ■ What a sheaf neural network is

A graph neural network (GNN) updates each node from its neighbours. The standard instance, the graph convolutional network (GCN), takes a forward-Euler step of the heat equation, then applies a weight matrix W and a pointwise nonlinearity σ [4]: with Δ_0 the normalized graph Laplacian,

$$\mathbf{X} \mapsto \sigma \left(\underbrace{(I - \Delta_0)}_{\text{Euler step}} \mathbf{X} W \right).$$

Heat flow averages neighbours, so it struggles when adjacent nodes ought to *differ*, and it washes features out as layers are stacked.

A **sheaf Laplacian** is the graph Laplacian with one change. Put $\mathbb{R}^d$ at each node and a matrix $\mathcal{F}_{v \trianglelefteq e}$ on each (node, edge) pair; to compare x_u and x_v , map both into a common space:

$$\langle \delta x, e \rangle = \mathcal{F}_{v \trianglelefteq e} x_v - \mathcal{F}_{u \trianglelefteq e} x_u, \quad L_{\mathcal{F}} = \delta^\top \delta.$$

With $d = 1$ and all matrices 1, this is the graph Laplacian. In general $\ker L_{\mathcal{F}}$ has dimension up to d rather than 1 — fix one fibre, propagate along a spanning tree — so diffusion can separate classes rather than collapse them. **Neural sheaf diffusion** [4] learns the matrices; [5,6] compute them; [7,8] fix them on meshes.

A vector space at each vertex and an invertible map along each edge is a discrete vector bundle with connection.

2 ■ What we claim

The models of [4,5,6] use degree at most one. The framework of [3] defines all degrees, but its discrete vector bundles are flat by construction, so curvature cannot arise there (block 8).

In degrees two and above we construct **curvature, a Bianchi identity, and products** on simplicial complexes, with no metric and no choice of coordinates.

Our subject is degree two and above.

3 ■ The object

A discrete vector bundle with connection on a locally ordered simplicial complex X is two pieces of data and one condition:

- ▶ a finite-dimensional vector space E_i for each vertex i ;
- ▶ an invertible linear map $U_{ji}: E_i \rightarrow E_j$ for each edge $[i, j]$, with $U_{ij} = U_{ji}^{-1}$.

Index order: the first index is the target. No metric, no embedding, no discretization map — the discrete forms are *given*. And $U_{ij} \in GL_n$: orthogonality is an extra assumption the theory does not need.

No metric is assumed, and nothing is attached to edges except the transport.

4 ■ A dictionary

In geometric ML	Here
stalk $\mathcal{F}(v)$; node feature space	fibre E_i , a vector space per vertex
restriction map $\mathcal{F}_{v \trianglelefteq e}$; learned edge map; $O(d)$ transport	parallel transport $U_{ji}: E_i \rightarrow E_j$, with $U_{ij} = U_{ji}^{-1}$
node signal $x \in \bigoplus_v \mathcal{F}(v)$	section $s \in C^0(X; E)$
sheaf coboundary δ	discrete covariant derivative $\langle \nabla s, [i, j] \rangle_i = U_{ij} s_j - s_i$
“loop matrix”; holonomy round a cycle	discrete curvature $F_{012} = U_{01} U_{12} - U_{02}$
trivial sheaf $\Rightarrow$ ordinary GCN	trivial bundle $\Rightarrow d_\nabla = d$, i.e. plain DEC [2]
sheaf Laplacian $\delta^\top \delta$; transports in $O(d)$	fibre metric with each U_{ij} an isometry: a <i>metric-compatible</i> connection [1]
no counterpart yet	$d_\nabla: C^k(X; E) \rightarrow C^{k+1}(X; E)$ for all k

Only the last row has no counterpart.

II. THE THEORY what sits above it

5 ■ Degrees above one

where current sheaf models operate

Each arrow is one transport applied to the single face that omits the origin vertex:

$$\langle d_\nabla \alpha, [0 \dots k] \rangle_0 = U_{01} \langle \alpha, [1 \dots k] \rangle_1 + \sum_{i=1}^k (-1)^i \langle \alpha, [0 \dots \widehat{i} \dots k] \rangle_0$$

Set every $U_{ij} = \text{Id}$ and this is the simplicial coboundary. Leibniz, naturality, $d_\nabla d_\nabla \alpha = F_\nabla \smile \alpha$ and Bianchi all hold, **with defining formulas identical to the smooth ones**; on a trivial bundle everything reduces to DEC [2].

F_∇ measures the failure of the sequence to be a cochain complex.

6 ■ Curvature is a holonomy defect

two ways of transporting E_2 into E_0

$$F_{012} = \langle F_\nabla, [0, 1, 2] \rangle_{0,2} = U_{01} U_{12} - U_{02} : E_2 \longrightarrow E_0$$

Curvature is a $\text{Hom}(E)$ -valued discrete 2-form, and it is exactly $d_\nabla \circ d_\nabla$. Transporting the value back to E_2 exhibits the holonomy with no choice of gauge:

$$U_{20} F_{012} = U_{20} U_{01} U_{12} - \text{Id} : E_2 \longrightarrow E_2,$$

the holonomy round $2 \rightarrow 1 \rightarrow 0 \rightarrow 2$ minus the identity — the Wilson-loop expression of lattice gauge theory, and the leading term of $\int F$ over a small square. Curvature obeys the Bianchi identity

$$d_\nabla^{\text{Hom}} F_\nabla = 0.$$

Curvature costs one $d \times d$ product per triangle and is gauge-covariant.

7 ■ What it costs

Let $N_k = |X_k|$ be the number of k -simplices, d the fibre dimension, f the channel count.

- ▶ d_∇ on $C^{k-1} \rightarrow C^k$ costs $O(N_k d^2 f)$ — the same order as one sheaf-diffusion layer.
- ▶ Forming F_{012} on every triangle costs $O(N_2 d^3)$. For its *action* on a section, $\langle F \smile s, [0, 1, 2] \rangle_0 = F_{012} s_2 = U_{01}(U_{12} s_2) - U_{02} s_2$, so F need never be materialized: $O(N_2 d^2 f)$.

On a mesh every N_k is $O(N_0)$. On a sparse graph the 2-cells are the graph's own triangles, $N_2 \leq \frac{1}{3} \sum_v \binom{\deg v}{2}$, enumerated once offline.

Degree 2 suffices. The expensive high-dimensional complexes are never required.

III. WHAT FOLLOWS and what is still open

8 ■ Why cellular sheaves are flat

Hansen and Ghrist note that for a discrete vector bundle “flatness is manifest in the fact that $\delta^2 = 0$ ” [3]. The reason is the functoriality axiom.

A cellular sheaf is a functor on the face poset: for $v \trianglelefteq e \trianglelefteq \tau$, $\mathcal{F}_{v \trianglelefteq \tau} = \mathcal{F}_{e \trianglelefteq \tau} \circ \mathcal{F}_{v \trianglelefteq e}$. When the maps are invertible, transport across *any* edge of a triangle τ equals

$$P_{u \rightarrow v} = \mathcal{F}_{v \trianglelefteq \tau}^{-1} \mathcal{F}_{u \trianglelefteq \tau},$$

independent of the edge, so composing round $\partial \tau$ telescopes to $P_{2 \rightarrow 0} P_{1 \rightarrow 2} P_{0 \rightarrow 1} = \text{Id}$: zero holonomy on every triangle. On a graph there are no 2-cells, so this is vacuous: nothing constrains the learned transports to be path-independent.

We have no transport from a vertex *directly* to a triangle, so $\mathcal{F}_{v \trianglelefteq \tau}$ does not exist and the composition condition has no left-hand side. This is not about impoverished data: under our coarsening every simplex carries a space, yet transports still join only adjacent dimensions, and $F_\nabla \neq 0$ survives [1, v2 §11.6].

General cellular sheaves also allow non-invertible maps and stalks of differing dimension; invertibility forces constant rank.

Curvature survives because no transport runs from a vertex straight to a triangle.

9 ■ Curvature obstructs anti-symmetrization

In scalar DEC the wedge product is an anti-symmetrized cup product [2]. For bundle-valued discrete forms that construction fails, and the obstruction is curvature itself.

On a tetrahedron with $\alpha \in C^1(X; E)$ and $w \in C^1(X)$, the Leibniz rule for the permutation $[0, 2, 3, 1]$ holds

$$\text{if and only if } U_{01} U_{12} = U_{02},$$

that is, iff the curvature on $[0, 1, 2]$ vanishes. Half the permutations carry such an obstruction, and they do not cancel.

Anti-symmetrization assumes flatness. The cup product does not.

10 ■ Why this is an Erlangen story

Define a geometry by the transformations preserving it. Two layers here, both theorems rather than design choices:

- ▶ **Local symmetry.** A basis in each fibre is a gauge; $A \mapsto g A g^{-1} - dg g^{-1}$ and $F \mapsto g F g^{-1}$.
- ▶ **Naturality.** $f^* d_\nabla = d_{f^* \nabla} f^*$ for every order-preserving simplicial map f .

Regarding a group G as a one-object category, equivariance is exactly naturality with respect to G : naturality restricted to *isomorphisms*. We have it for all morphisms, invertible or not.

The maps a network applies are mostly not invertible: restriction to a region, the sampled subgraph of **minibatching**. Naturality covers them, and “order-preserving” costs nothing — the subdivision functor sends any simplicial map f to an order-preserving $\text{sd} f: \text{sd} \tilde{X} \rightarrow \text{sd} \tilde{Y}$. Equivariance under a subgroup of $\text{Aut}(X)$ constrains operators on X alone.

Naturality relates operators on two different complexes. Equivariance only ever speaks about one.

11 ■ Open problems — please come and talk

- ▶ **Hodge theory without flatness.** Both metrics are in hand: fibre metrics give metric-compatible connections ($U_{ij} \in O(n)$), and the base metric ports from DEC through the circumcentric subdivision. What fails is the *decomposition*, which needs $d_\nabla \circ d_\nabla = 0$: curvature obstructs Hodge theory too.
- ▶ **Is curvature a useful learned feature?** F_{012} is local, gauge-covariant and cheap. As far as we know it has not been measured on a trained model.
- ▶ **Gauge fixing is parameter reduction.** On a connected graph with n vertices and m edges only $\beta_1 = m - n + 1$ holonomies are observable; a spanning-tree gauge removes the other $n - 1$ matrices, and a flat direction of the loss with them.
- ▶ **A PhD research assistantship** is open, starting Summer 2027.

12 ■ Three things to take away

1. The sheaf models and this theory describe the **same object**.
2. Above degree one it has **curvature**: one $d \times d$ matrix per triangle. It obstructs anti-symmetrizing the cup product, and it stops the Hodge decomposition, which needs $d_\nabla \circ d_\nabla = 0$.
3. A cellular sheaf on a complex of dimension ≥ 2 is **flat by axiom**, so the sheaf framework cannot reach it.

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